HealthLedger AI

Patent, Trademark, Trade Dress & Copyright Disclosure

AI-powered accounting compliance for hospital finance departments. Custom-trained on cooperation agreements. Grounded in doctoral research. Built from the inside — through on-site immersion.

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Brand Name:	**HealthLedger AI**
Domain:	**healthledgerai.com**
Contact Email:	**info@healthledgerai.com**
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A. BRAND IDENTITY & INTELLECTUAL PROPERTY OVERVIEW

This disclosure covers four distinct categories of intellectual property arising from the HealthLedger AI project. Patent counsel is advised to evaluate each category and advise on the appropriate registration strategy in each applicable jurisdiction.

IP Category	Subject Matter	Recommended Action
Patent	AI system, RAG pipeline, 3-step method, on-site methodology, UI innovations	Provisional application then PCT/national phase
Trademark	Name 'HealthLedger AI', domain, taglines, logo dot	TM registration in DE/EU/US (EUIPO + USPTO)
Trade Dress	Colour palette, botanical mark, diamond ornament	Include in TM application as design mark
Copyright	All source code, SVG artwork, CSS design system	Automatic; register with copyright office for damages
Trade Secret	On-site immersion methodology details, instit. training data	NDA / confidentiality agreements with partners

A.1 Trademark Disclosure — Name, Domain & Taglines

The following marks are claimed for trademark registration:

Primary Mark:	**HealthLedger AI**
Class (NICE):	**Class 42 — Software as a service; AI services; compliance software for healthcare**
Domain in Use:	**healthledgerai.com (CNAME record confirmed in repository)**
Contact Email:	**info@healthledgerai.com**
Nav Subtitle:	**"Healthcare Accounting Intelligence" — secondary mark candidate**
Hero Tagline:	**"The intelligence healthcare accounting has been waiting for."**
Footer Tagline:	**"Something important is being built." — brand statement**
Logo Element:	**Pulsing sage-green dot (animated) adjacent to logotype text**
First Use Date:	**2026 (per copyright notice on all source files and website)**
R&D; Statement:	**"Research & Development — 2026" (shown in hero eyebrow)**

NOTE TO COUNSEL: 'HealthLedger' and 'HealthLedger AI' should both be filed. The domain healthledgerai.com establishes first use in commerce. The email domain info@healthledgerai.com provides further evidence of commercial operation. Consider filing in EUIPO (EU-wide), USPTO (US), and DPMA (Germany) simultaneously. The tagline 'Something important is being built.' and the subtitle 'Healthcare Accounting Intelligence' are strong secondary mark candidates.

A.2 Trade Dress Disclosure — Visual Identity

The following visual elements form a distinctive and consistent trade dress. The combination as a whole creates a non-functional aesthetic identity strongly associated with HealthLedger AI:

Cream #F4EFE4	Primary background — warm parchment. Dominant surface colour.
Gold #A08558	Primary accent. Used for all key highlights, rules, and borders.
Sage #4A6B5A	Secondary accent. Logo dot, botanical, italic headline words.
Ivory #EDE7D8	Secondary surface. Contact section background.
Blush #D4A9A0	Tertiary. Rare accent, particle field only.
Ink #1E1A14	Near-black warm. All primary text.

- **Botticelli-Inspired Lily SVG:** An original hand-crafted SVG botanical illustration depicting a lily with three blooms, three leaves, and a curved stem. Rendered in sage green (#4A6B5A) with ivory/cream petals (#EDE7D8, #F4EFE4) and gold stamens (#A08558). Displayed at 13% opacity as a fixed background element with parallax scrolling. This illustration is original artwork and is both a copyright work and a trade dress element.
- **Diamond Ornament:** A 45°-rotated square (SVG rect, rotated 45°) used as a repeating geometric motif — in floating ornaments, section dividers, and step number accents. Rendered in gold (#C9AA7C). This diamond shape appears consistently throughout the interface as a brand signature.
- **Pulsing Logo Dot:** An 8px sage (#4A6B5A) circle with a breathing box-shadow pulse animation (2.8s cycle) positioned immediately left of the logotype. This animated element functions as the logo mark.
- **Typography Pairing:** Cormorant Garamond serif (headings, logotype) paired with Jost sans-serif (body, labels). Weights 300–500 only. This specific pairing is a distinctive element of the brand.
- **Gold Hairline Rules:** 1px gold (#C9AA7C) horizontal rules used as section dividers throughout the interface, including nav borders and footer.
- **Grain Texture Overlay:** SVG fractalNoise filter (baseFrequency 0.9, 4 octaves) at 4% opacity applied as a fixed full-screen texture, giving a tactile parchment quality to all surfaces.
- **Radial Gradient Ambient Layer:** Three overlapping radial gradients (gold, sage, blush at 5–12% opacity) creating a painterly atmospheric depth effect.
- **Dark Intro Backdrop:** #15120E (warm near-black) — used exclusively for the intro animation, creating a dramatic before/after contrast with the cream palette.

A.3 Copyright Notice — All Source Files

All source code, artwork, and design assets in this disclosure are original works of authorship created by Maria Polychroniadou. Copyright subsists automatically upon creation. The following notice applies to all files:

Files Covered:	managed-agents-starter.ts, app/page.tsx, app/globals.css, app/layout.tsx, next.config.js, components/Botanical.tsx, components/CustomCursor.tsx, components/FloatingOrbs.tsx, components/HeroScene.tsx, components/IntroOverlay.tsx, components/Marquee.tsx, components/ParticleField.tsx, components/ProblemHook.tsx, components/SectionContact.tsx, components/SectionHow.tsx
Artwork:	SVG Lily Botanical (Botanical.tsx) — original vector artwork
Design System:	CSS custom property palette (globals.css) — original design
Repository:	mrply97/AI (GitHub) — git history establishes creation dates

NOTE TO COUNSEL: Registration of copyright (with the relevant copyright office) is recommended to enable statutory damages in infringement actions. The git repository history provides timestamped evidence of creation for all files.

1. FIELD OF THE INVENTION

The present invention relates to artificial intelligence systems for healthcare financial administration and compliance. More particularly, the invention concerns: (i) a managed AI agent system trained on hospital cooperation agreement corpora and deployed as a real-time clause-traceable compliance interface for hospital accounting departments; (ii) a novel on-site immersion methodology for capturing institutional accounting knowledge as AI training data; and (iii) a suite of interactive user interface components implementing a distinctive branded compliance experience, including animated particle systems, 3D parallax rendering, and a constellation particle field.

2. BACKGROUND OF THE INVENTION

The following problems, identified through doctoral research and direct on-site engagement with hospital finance teams, constitute the background art against which the invention must be assessed:

- **Volume and Complexity of Cooperation Agreements.** A single hospital network maintains in excess of 200 active cooperation agreements with insurers, pharmaceutical suppliers, device manufacturers, and other healthcare entities. Each agreement contains clauses with direct accounting obligations: revenue recognition triggers, cost allocation rules, reconciliation requirements, and compliance thresholds. Across a hospital network, this produces several thousand individually actionable accounting obligations.
- **Inconsistent Interpretation Creates Financial Exposure.** Without a machine-readable authoritative source, cooperation agreement clauses are interpreted differently by different accounting staff. A single misread clause generates liability of €10,000–€500,000+ per fiscal period. This problem is structural and persists across all hospital accounting departments operating under German, Austrian, Swiss, and EU healthcare finance law.
- **No Domain-Specific AI Solution Exists.** General-purpose AI assistants (OpenAI ChatGPT, Google Gemini, Anthropic Claude in its default configuration) are not trained on hospital cooperation agreement corpora, do not know the institution's chart of accounts, and cannot produce clause-traceable, auditable compliance determinations. No commercial product addresses this gap.
- **Tacit Knowledge is Irreplaceable and Inaccessible.** Hospital accounting teams accumulate interpretive conventions — informal precedents and team-specific practices — that are never written down. No prior art system captures this knowledge. The present invention's on-site immersion methodology is the only known approach to converting this tacit knowledge into AI training material.
- **The Problem is Grounded in Doctoral Research.** The inventor's original research provides the evidence base for the problem statement and the quantification of financial exposure. This academic grounding constitutes a further distinguishing feature of the invention's provenance.

3. SUMMARY OF THE INVENTION

HealthLedger AI is a multi-component invention comprising: (A) a core AI compliance system (the primary patent subject matter); and (B) a suite of novel user interface innovations (secondary patent subject matter). Both components share the HealthLedger AI brand identity (trademark subject matter) and the distinctive visual system (trade dress subject matter).

A. Core AI Compliance System — Four Pillars

Pillar I — Cooperation Agreement Mapping
Automated ingestion, semantic segmentation, obligation-type classification, vector embedding, and indexed storage of hospital cooperation agreement clauses. Introduces a six-type healthcare accounting clause taxonomy (RevenueRecognition, CostAllocation, ComplianceTrigger, RevenueThreshold, PenaltyClause, ReconciliationRule).

Pillar II — Accounting Compliance Intelligence
Retrieval-augmented generation layer: on receipt of an accounting query, retrieves the top-K semantically closest clauses, assembles an augmented prompt, and generates a compliance determination citing exact clause IDs and parent agreement names — producing an auditable, defensible determination.

Pillar III — Hospital-Specific AI Model
Persistent managed AI agent entity carrying institution-specific instructions assembled from: the hospital's chart of accounts; internal naming convention mappings; and historical interpretive decisions captured on-site. Agent identity is stable; individual queries are mediated through discrete, ephemeral sessions.

Pillar IV — On-Site Immersion Methodology
A novel AI development methodology: sustained on-site engagement with hospital accounting teams to observe accounting decisions, elicit tacit interpretive conventions, and produce ground-truth training data unavailable from any other source. Constitutes an independently patentable method.

B. User Interface Innovations — Novel Interactive Components

- **Timed Particle Burst Intro Animation:** A four-phase animated sequence (IDLE 2.6s, BURST 5.4s, HOLD 6.4s, FADE 7.8s) using a diamond-glitter particle system on a dark backdrop, transitioning to the cream-palette main interface.
- **3D Parallax Hero Scene with Spring Physics:** Mouse-reactive 3D tilt of the entire hero canvas using spring physics (stiffness 70, damping 18, mass 0.4), with depth-differentiated child layers creating a spatial parallax.
- **Brand-Palette Constellation Particle Field:** Interactive particle network using brand colours (gold, sage, blush) with dual-mode physics: Lissajous-pattern organic drift and mouse-attraction field (radius 180px).
- **Dual-Spring Branded Cursor:** Two-component cursor system: fast inner dot (gold, stiffness 600) and slow outer ring (stiffness 100), both spring-animated to the mouse position with independent damping.
- **Botanical SVG Illustration:** Original Botticelli-inspired lily artwork in SVG, integrated as a 3D parallax layer with scroll-driven drift animation.

■ **Sequential Problem-Hook Display:** Three-line problem statement rotated at 1,900ms intervals using AnimatePresence — a distinctive UX framing device.

■ **Word-Flip Kinetic Headline:** Headline words revealed individually via 3D rotateX flip animation (90°→0°) with staggered delays from t=6.2s.

4. DETAILED TECHNICAL DESCRIPTION

4.1 System Architecture — Five-Layer Model

Layer 1 — Document Knowledge Base

Vector database (Pinecone / Weaviate / pgvector) storing 1536-dimensional embeddings of all agreement clauses, indexed by clauseId, agreementId, clauseType, effectiveDate. Supports approximate nearest-neighbour search for semantic clause retrieval.

Layer 2 — AI Agent Kernel

Persistent managed AI agent entity (Anthropic beta.agents.create) with model claude-opus-4-8. Agent holds institution-specific instructions. Agent ID is stored durably and reused for all sessions — never recreated after initial setup per institution.

Layer 3 — Session Orchestration

Session layer (Anthropic beta.sessions.create / events.stream) mediates each user query as an ephemeral stateful session against the persistent agent. Retrieval-augmented prompt is assembled and injected per session.

Layer 4 — Compliance Output Module

Post-processor extracts: compliance determination text; cited clause IDs (pattern [AGR-XX §Y.Z]); parent agreement names; confidence indicator. Writes structured record to compliance audit log.

Layer 5 — Frontend Application

Next.js 14 static export (output: 'export') deployed to healthledgerai.com via GitHub Pages / CNAME record. React 18 client components with framer-motion animations. Provides compliance UI, research enrolment portal, and on-site knowledge capture interface.

4.2 AI Agent Module & Session Architecture

The Agent Module implements a create-once / reuse-always pattern. On institution onboarding, agents.create() is called with the institution-specific instruction set. The returned agent.id is stored in a persistent data store. Every subsequent accounting query creates a new session (sessions.create({ agent_id })) referencing this stored ID. Response tokens are streamed via server-sent events (events.stream), enabling first-token display in under 800ms. This architecture maintains institutional knowledge in the long-lived agent while keeping per-query state isolated in ephemeral sessions.

4.3 Agreement Ingestion & RAG Pipeline — Six-Step Process

Step 1: PDF parsing → plain text (preserve section numbers as metadata)

Step 2: NLP clause segmentation (boundary detection on structural cues)

Step 3: Clause-type classification → one of 6 ClauseType values

Step 4: Vector embedding → 1536-dimensional representation per clause

Step 5: Vector DB upsert with metadata (clauseId, agreementId, effectiveDate)

Step 6: RAG query: embed user question → ANN search → top-5 clause retrieval → prompt assembly

4.4 Domain Training & Institutional Knowledge Module

Three knowledge sources are assembled into the agent instruction set: (1) the hospital's chart of accounts (code → name mapping); (2) naming convention dictionary (internal term → standard term); (3) historical InterpretationRecords (clauseRef, decision, rationale, date) captured during on-site immersion. The buildInstructions() function assembles these into a structured agent instruction string. Updates are applied incrementally via agent instruction refresh — no model retraining required.

4.5 Live Compliance Engine

Query lifecycle: (1) User submits question via web UI. (2) Query is embedded. (3) Vector DB returns top-5 semantically similar clauses. (4) Augmented prompt assembled: clauses as [AGR-ID §clause-ID]: text format + query + cite-clause instruction. (5) Session created; prompt submitted to agent. (6) Response streamed token-by-token (<800ms first-token target). (7) Response post-processed to extract: determination, clause citations, agreement references. (8) Audit log entry written (query, response, citations, timestamp, user).

4.6 Animated Entry Experience — IntroOverlay Component

The IntroOverlay implements a four-phase diamond-glitter particle animation that functions as the system's entry experience:

Phase 1 — IDLE (0–2,600ms):

Up to 950 particles arranged on a unit sphere using cubic-root radial distribution for uniform density. Sphere rotates on Y-axis (rate: elapsed×0.00035 rad/ms) with sinusoidal X tilt. 16% of particles are designated 'glints' with eighth-power sparkle pulses. Colour palette: silver-white (cool, contrasting with brand palette).

Phase 2 — BURST (2,600–5,400ms):

Sphere radius expands from baseRadius (16% of min viewport dimension) outward by factor ×9 using easeInOutCubic interpolation. Particles fly outward maintaining their direction vectors from the unit sphere positions.

Phase 3 — HOLD (5,400–6,400ms):

Scattered glitter pattern holds static on dark (#15120E) background.

Phase 4 — FADE (6,400–7,800ms):

Dark overlay root opacity transitions 1→0 using easeInQuad, revealing the cream-palette main interface beneath. Animation frame loop terminates at 7,800ms; component unmounts (setDone(true)).

4.7 Interactive Particle Field — ParticleField Component

A full-viewport canvas particle system rendering up to 120 particles (density: floor(w×h/9500)) in the brand colour palette. Two physics modes operate simultaneously: (1) Lissajous organic drift — vx += sin(t×0.9 + phase)×0.005 / vy += cos(t×0.7 + phase×1.4)×0.004; (2) mouse attraction — pull force = ((180–dist)/180)^1.8 × 0.014 toward cursor. Damping: ×0.955 per frame. Particles render with a radial glow corona (radius×3) plus a solid core. Constellation lines (rgba(160,133,88,...)) drawn between particles within 130px; bright lines drawn from particles to cursor within 180px.

4.8 3D Parallax Scene & Floating Diamond Ornaments

HeroScene wraps all hero content in a CSS 3D perspective container (perspective: 1200px, origin: 50% 40%). Mouse movement drives two spring-animated rotation values (rotateY: ±7°, rotateX: ±5°, spring params: stiffness 70 / damping 18 / mass 0.4). Child elements at different translateZ values create natural parallax depth.

FloatingOrbs renders 7 rotating-square diamond SVG ornaments at z-depths 30–90px (closer = more parallax movement), at positions distributed in the right 70–91% and left 6–10% of the viewport. All appear after a 8-second delay (post headline animation) with opacity cycling [0, 0.55, 0.35, 0.55] and individual float periods of 6.5–9.8s. Two gold gradient horizontal accent lines at translateZ 35px and 55px pulse with scaleX animation.

4.9 Branded Cursor System — CustomCursor Component

On fine-pointer devices (mouse), the system cursor is hidden and replaced by two spring-animated elements: an inner gold dot (7px, fill #A08558, box-shadow 0 0 8px rgba(160,133,88,0.6), spring stiffness 600 / damping 35) and an outer ring (34px diameter, border 1px solid rgba(160,133,88,0.45), mix-blend-mode: multiply, spring stiffness 100 / damping 20). The dual-spring creates a trailing lag between dot and ring that is a distinctive brand interaction.

4.10 Botanical SVG Illustration — Original Artwork

The Botanical component renders an original SVG lily illustration (viewBox 300×600) comprising: a curved bezier stem (stroke #4A6B5A, width 2); three organic leaf shapes at y≈300, 380, 420 (fill #4A6B5A, opacity 0.5–0.7); a full lily bloom at translate(148,80) with five petals (cream/ivory fill, gold-lt stroke) and three stamens terminating in gold circles; a side bloom at translate(178,200) rotate(30°); and a bud at translate(125,160) rotate(-15°). The artwork is positioned as a 3D parallax layer (translateZ -100px, scale 1.1) with scroll-driven upward drift (0%→-22% over 700px). The illustration is an original copyright work and a trade dress element.

4.11 Frontend Architecture & Deployment

Framework: Next.js 14 (React 18), TypeScript, framer-motion 11. Build output: static HTML export (output: 'export' in next.config.js). Deployment: GitHub Pages via CNAME record pointing healthledgerai.com to the repository. Typography: Cormorant Garamond (serif, weights 300–600, italic variants) + Jost (sans-serif, weights 200–500) loaded via Google Fonts. Metadata title: 'HealthLedger AI — Accounting Intelligence for Healthcare'. Metadata description: 'AI-powered compliance built specifically for hospital accounting departments. Custom-trained on cooperation agreements. Grounded in doctoral research.' The metadata description is a registered brand statement.

4.12 Data Flow & API Integration

Data	From	To	Protocol
Agreement PDFs	Hospital IT/Legal	Ingestion pipeline	HTTPS upload
Clause embeddings	Embedding model	Vector database	API call
On-site capture records	Immersion tool	Institution DB	HTTPS POST
Updated instructions	Domain trainer	Anthropic agent	agents API
Accounting query	Accountant (web)	RAG engine	HTTPS POST

Augmented prompt	RAG engine	Agent session	sessions API
Streamed response	Agent session	Accountant UI	SSE stream
Audit log entry	Output module	Compliance DB	DB write
Research enrolment	SectionContact UI	info@healthledgerai	mailto link

5. SOURCE CODE LISTINGS

All listings below are original works © 2026 HealthLedger AI — Maria Polychroniadou. Production code is reproduced verbatim from the repository. Annotated pseudocode represents inventive modules under active development.

5.1 Core AI Agent Module

Listing 5.1 — managed-agents-starter.ts (production, verbatim)

```
// FILE: managed-agents-starter.ts | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Core AI Agent Orchestration Module
// NOVELTY: Persistent managed-agent identity + per-query session pattern for
// healthcare accounting compliance.

import Anthropic from "@anthropic-ai/sdk";
const client = new Anthropic(); // ANTHROPIC_API_KEY from environment

// ■■ AGENT CREATION (run once per institution onboarding) ■■■■■■■■■■■■■■■■■■■■■■■■■■
async function createAgent() {
const agent = await client.beta.agents.create({
name: "healthledger-assistant", // Institution-scoped agent identity
model: "claude-opus-4-8", // Large-scale transformer (LLM)
instructions:
"You are the HealthLedger AI compliance assistant for [INSTITUTION_NAME]. " +
"Answer accounting compliance questions by reasoning over cooperation agreement " +
"clauses. Always cite the exact clause ID and agreement reference in your answer.",
});
console.log(`AGENT_ID=${agent.id}`); // Store durably — never recreate
return agent;
}

// ■■ SESSION EXECUTION (called per accounting compliance query) ■■■■■■■■■■■■■■■■■■■■
async function runSession(agentId: string, augmentedPrompt: string) {
const session = await client.beta.sessions.create({ agent_id: agentId });

// Token-by-token streaming → real-time display in accounting UI (< 800ms TTF)
const stream = client.beta.sessions.events.stream(session.id, {
input: [{ role: "user", content: augmentedPrompt }],
});

for await (const event of stream) {
if (event.type === "content_block_delta" && event.delta.type === "text_delta") {
process.stdout.write(event.delta.text); // Pipe to UI
}
}
return session;
}

async function main() {
const agentId = process.env.AGENT_ID;
if (!agentId) { await createAgent(); return; }

// Example: augmented prompt assembled by RAG engine (see Listing 5.2)
const prompt = `
Retrieved clauses from Agreement #47 (MedTech GmbH, eff. 2025-01-01):
[AGR-47 §3.2]: "Revenue from device maintenance services shall be recognised
in the period in which the service obligation is fulfilled..."
[AGR-47 §3.5]: "Quarterly reconciliation required within 30 days of period close."
```

```
Accounting question: How do we recognise Q3 maintenance revenue from MedTech?
Cite the exact clause. Use account code 4300 (Service Revenue).
`;
await runSession(agentId, prompt);
}
main().catch(console.error);
```

5.2 Agreement Ingestion & RAG Pipeline

Listing 5.2 — lib/ingestion-pipeline.ts (annotated pseudocode)

```
// FILE: lib/ingestion-pipeline.ts | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Agreement Ingestion & Retrieval-Augmented Generation Pipeline
// NOVELTY: Hospital-specific clause taxonomy + semantic retrieval for accounting compliance

interface AgreementClause {
clauseId: string; // e.g. "AGR-47-§3.2"
agreementId: string; // Parent agreement
text: string;
type: ClauseType;
obligations: string[];
effectiveDate: Date;
embedding?: number[]; // 1536-dim vector
}

enum ClauseType {
RevenueRecognition = "revenue", // When/how revenue is booked
CostAllocation = "cost", // How costs are split
ComplianceTrigger = "compliance", // Events requiring accounting action
RevenueThreshold = "threshold", // Volume/value gates
PenaltyClause = "penalty", // Financial penalties → liability
ReconciliationRule = "reconciliation" // Periodic settlement obligations
}

// STEP 1 — Document ingestion (called once per agreement)
async function ingestAgreement(pdfPath: string, agreementId: string) {
const rawText = await parsePDF(pdfPath); // PDF → plain text
const clauses = await segmentClauses(rawText); // NLP clause detection
const tagged = await tagClauseTypes(clauses); // Classify each clause
const embedded = await embedClauses(tagged); // text-embedding-ada-002 or equiv
await vectorDB.upsert(embedded, { agreementId }); // Index in Pinecone / pgvector
return embedded;
}

// STEP 2 — RAG query (called per user question)
async function queryCompliance(agentId: string, userQuery: string) {
const qEmbed = await embedQuery(userQuery); // Embed question
const topK = await vectorDB.search(qEmbed, { topK: 5 }); // ANN search
const context = topK.map(c =>
`[${c.agreementId} ${c.clauseId}]: ${c.text}`).join("\n\n");
const prompt = `Context:\n${context}\n\nQuestion: ${userQuery}\nCite clause.`;
const response = await runSession(agentId, prompt); // LLM reasoning

return {
determination: response.text,
sourceClauses: topK.map(c => c.clauseId),
agreements: topK.map(c => c.agreementId),
timestamp: new Date(),
};
}
```

5.3 Domain Training Module

Listing 5.3 — lib/domain-training.ts (annotated pseudocode)

```
// FILE: lib/domain-training.ts | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Institutional Knowledge Capture & Domain Training Module
// NOVELTY: On-site immersion methodology → agent instruction assembly

interface InstitutionProfile {
hospitalId: string;
name: string;
chartOfAccounts: { code: string; name: string }[];
namingConventions: Record<string, string>; // e.g. {"Servicevertrag" → "Service Agreement"}
interpretations: InterpretationRecord[];
}

interface InterpretationRecord {
clauseRef: string; // AGR-47-§3.2
decision: string; // "Recognise in Q3, account 4300"
rationale: string; // Expert justification (on-site captured)
capturedBy: string; // Finance team member name
capturedAt: Date;
}

// Capture a new on-site interpretive decision
async function captureInterpretation(hospitalId: string, record: InterpretationRecord) {
await db.interpretations.insert({ hospitalId, ...record });
await refreshAgentInstructions(hospitalId); // Incremental update, no retraining
}

// Assemble institution-specific agent instruction set from on-site data
async function buildInstructions(hospitalId: string): Promise<string> {
const p = await db.institutions.findById(hospitalId);
const h = await db.interpretations.findByHospital(hospitalId, { limit: 30 });

return `
You are HealthLedger AI for ${p.name}.

CHART OF ACCOUNTS — always use these codes:
${p.chartOfAccounts.map(a => ` ${a.code}: ${a.name}`).join("\n")}

NAMING CONVENTIONS:
${Object.entries(p.namingConventions).map(([k,v]) => ` "${k}" = "${v}"`).join("\n")}

PRECEDENT DECISIONS (from on-site immersion):
${h.map(r => ` [${r.clauseRef}] → ${r.decision} (${r.rationale})`).join("\n")}

Rules:
- Always cite the exact clause ID and agreement name.
- Use only the institution's account codes above.
- Flag ambiguous clauses explicitly.
- Do not extrapolate beyond the retrieved clauses.
`;
}
```

5.4 Main Application Page

Listing 5.4 — app/page.tsx (production, key constants)

```
// FILE: app/page.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Main application page — Next.js 14 client component

"use client";
import { motion, useScroll, useTransform } from "framer-motion";
import ProblemHook from "@/components/ProblemHook";
import IntroOverlay from "@/components/IntroOverlay";
import Botanical from "@/components/Botanical";
import HeroScene from "@/components/HeroScene";
import FloatingOrbs from "@/components/FloatingOrbs";
import ParticleField from "@/components/ParticleField";
import CustomCursor from "@/components/CustomCursor";
import Marquee from "@/components/Marquee";
import SectionHow from "@/components/SectionHow";
import SectionContact from "@/components/SectionContact";

// Brand-defining keyword set — used in marquee ticker
const MARQUEE_TEXT =
"Cooperation Agreements · Accounting Compliance · Hospital Finance · " +
"AI-Powered Intelligence · On-Site Research · Healthcare Accounting · " +
"Doctoral Research · Revenue Compliance · ";

// Four pillars — directly correspond to the four patent claims
const PILLARS = [
"Cooperation Agreement Mapping",
"Accounting Compliance",
"Hospital-Specific AI",
"On-Site Methodology",
];

// Problem statement — distilled from doctoral research, 1900ms interval
const PROBLEM_HOOKS = [
"A hospital has 200+ cooperation agreements.",
"Each interpreted differently by accounting.",
"One clause. Thousands in exposure.",
];

// Taglines (trademark candidates)
const HERO_LINES = [
[{ text: "The" }, { text: "intelligence", cls: "word-italic-sage" }],
[{ text: "healthcare" }, { text: "accounting" }],
[{ text: "has been" }, { text: "waiting for.", cls: "word-italic-gold" }],
];
const FOOTER_TAGLINE = "Something important is being built.";
const NAV_SUBTITLE = "Healthcare Accounting Intelligence";

// Brand identity constants
const BRAND_NAME = "HealthLedger AI";
const BRAND_DOMAIN = "healthledgerai.com";
const BRAND_EMAIL = "info@healthledgerai.com";
const COPYRIGHT_YEAR = 2026;
```

5.5 IntroOverlay & Particle Burst Animation

Listing 5.5 — components/IntroOverlay.tsx (production, annotated)

```
// FILE: components/IntroOverlay.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Animated particle-burst intro sequence
```

```
// NOVELTY: Timed four-phase diamond-glitter animation (brand-specific colour palette)
// that functions as a distinctive interactive entry experience.

// Timing constants (milliseconds) — define the proprietary animation sequence
const IDLE_END = 2600 // Dense spherical particle swarm breathes
const BURST_END = 5400 // Particles burst outward (easeInOutCubic)
const HOLD_END = 6400 // Scattered glitter holds on dark background
const FADE_END = 7800 // Dark overlay dissolves into the cream page

// Silver/diamond palette (intentionally distinct from brand gold — cool intro)
const INTRO_PALETTE = [
[255, 255, 255], // pure white
[228, 234, 242], // silver-white
[200, 212, 226], // light steel
[255, 250, 245], // warm glint
[180, 196, 214], // cool silver
];

// Sphere generation: cubic-root distribution for dense fill (not shell-only)
// count = min(floor(w*h / 2200), 950) — scales to viewport size
// Each particle has: unit-sphere position (bx,by,bz), size, colorIdx,
// twinklePhase, isGlint flag (16% probability), sparkleSpeed

// Render loop: uses Y-axis rotation (rotY += elapsed * 0.00035) and
// sinusoidal X-axis tilt for organic 3D breathing motion.
// isGlint particles get an 8th-power sparkle pulse for diamond effect.
// Background: #15120E (near-black warm dark) — dramatic brand contrast.
```

5.6 ParticleField — Constellation Network

Listing 5.6 — components/ParticleField.tsx (production, annotated)

```
// FILE: components/ParticleField.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Interactive constellation particle network
// NOVELTY: Brand-palette particle field with dual-mode physics
// (Lissajous organic drift + mouse attraction) and constellation lines.

// Brand colour palette used for particles:
const PARTICLE_PALETTE = [
[160, 133, 88], // gold #A08558
[201, 170, 124], // gold-lt #C9AA7C
[ 74, 107, 90], // sage #4A6B5A
[122, 158, 138], // sage-lt #7A9E8A
[212, 169, 160], // blush #D4A9A0 (rare — ~12% of particles)
];

// Physics constants:
const CONNECT_DIST = 130; // px — constellation line draw distance
const MOUSE_RADIUS = 180; // px — attraction field radius
const PARTICLE_COUNT = min(floor(w*h / 9500), 120); // density formula

// Per-particle state: position (x,y), velocity (vx,vy), size, alpha, color, phase
// Organic drift: vx += sin(t*0.9 + phase)*0.005; vy += cos(t*0.7 + phase*1.4)*0.004
// Mouse pull: if dist < 180: vx += (dx/dist) * pull^1.8 * 0.014
// Damping: vx *= 0.955; vy *= 0.955 (inertial feel)
// Radial glow: createRadialGradient(p.x, p.y, 0, p.x, p.y, size*3)
// Constellation: strokeStyle rgba(160,133,88, (1-d/130)*0.14) — gold lines
// Cursor lines: opacity = (1 - dist/180)^1.5 * 0.45 — gold to cursor
```

5.7 HeroScene · FloatingOrbs · CustomCursor

Listing 5.7 — HeroScene / FloatingOrbs / CustomCursor (annotated)

```
// FILE: components/HeroScene.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: 3D Parallax Tilt Scene Container
// NOVELTY: Spring-physics 3D tilt with depth-layered child elements
// creating a distinctive interactive spatial perspective effect.

// Spring parameters (tuned for brand feel):
const TILT_Y = useSpring(useTransform(rawX, [-0.5, 0.5], [-7, 7]),
{ stiffness: 70, damping: 18, mass: 0.4 });
const TILT_X = useSpring(useTransform(rawY, [-0.5, 0.5], [ 5, -5]),
{ stiffness: 70, damping: 18, mass: 0.4 });

// Perspective: 1200px from scene-root; origin at 50% 40% (slightly above centre)
// Child z-depths:
// .layer-ambient → translateZ(-160px) (background, moves least)
// .botanical-layer → translateZ(-100px) (lily illustration, subtle parallax)
// FloatingOrbs → translateZ(30–90px) (diamonds, move most — foreground)
// .scene-content → translateZ(0) (text, neutral depth)

// FILE: components/FloatingOrbs.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Floating diamond ornament layer
// 7 rotating-square (45°) diamond SVG ornaments at varying z-depths and positions
// 2 thin gold gradient horizontal accent lines
// All appear AFTER headline animation (delay = 8s) to not compete with text
// Opacity cycling: [0, 0.55, 0.35, 0.55] with individual drift durations 6.5–9.8s

// FILE: components/CustomCursor.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: Dual-spring branded cursor
// Inner dot: 7px gold (#A08558) circle, fast spring (stiffness:600, damping:35)
// box-shadow: 0 0 8px rgba(160,133,88,0.6)
// Outer ring: 34px gold-border circle, slow spring (stiffness:100, damping:20)
// border: 1px solid rgba(160,133,88,0.45), mix-blend-mode: multiply
// Active only on fine-pointer devices (mouse); hidden on touch/mobile
```

5.8 Botanical SVG Illustration & Marquee Ticker

Listing 5.8 — Botanical.tsx / Marquee.tsx (annotated)

```
// FILE: components/Botanical.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
// DESCRIPTION: SVG Lily Illustration — Original Artwork
// NOVELTY: Hand-crafted SVG botanical illustration in Botticelli tradition,
// rendered as a React component with 3D parallax and scroll animation.
//
// SVG viewBox: 0 0 300 600
// Stem: bezier M150 580 C148 480 ... C160 220 150 160 148 80
// stroke #4A6B5A (sage), strokeWidth 2, rounded caps
// Leaves: 3 organic bezier leaf shapes at y≈300, 380, 420
// opacity 0.5–0.7, fill #4A6B5A
// Bloom 1 (full): 5 petals at translate(148,80)
// petal paths: cream fill #EDE7D8 / ivory fill #F4EFE4
// stroke: #C9AA7C (gold-lt), strokeWidth 0.8
// stamens: 3 lines to #A08558 (gold), tip circles r=3 fill #C9AA7C
// Bloom 2 (side, 4 petals): translate(178,200) rotate(30)
// Bud: translate(125,160) rotate(-15), 3 petal paths, opacity 0.7
//
// Parallax: translateZ(-100px) within 3D scene; scale(1.1) compensates shrinkage
// Scroll: MotionValue scrollY → y: 0% → -22% over first 700px of scroll

// FILE: components/Marquee.tsx | © 2026 HealthLedger AI — Maria Polychroniadou
```

```
// Brand keyword ticker — animation: marquee-scroll 30s linear infinite
// Text: "Cooperation Agreements · Accounting Compliance · Hospital Finance ·
// AI-Powered Intelligence · On-Site Research · Healthcare Accounting ·
// Doctoral Research · Revenue Compliance · "
// Font: Jost 11px, weight 400, letter-spacing 0.18em, uppercase, color #8A8070
// Border: rgba(160,133,88,0.18) top and bottom — gold hairline
```

5.9 Design System — globals.css

Listing 5.9 — app/globals.css (production, key extracts)

```
/* FILE: app/globals.css | © 2026 HealthLedger AI — Maria Polychroniadou */
/* DESCRIPTION: Complete visual design system — CSS custom properties */

:root {
/* ■■ Brand Colour Palette ■■ */
--cream: #F4EFE4; /* Primary background — warm parchment */
--ivory: #EDE7D8; /* Secondary surface — aged paper */
--gold: #A08558; /* Primary accent — antique gold */
--gold-lt: #C9AA7C; /* Light gold — borders, lines */
--sage: #4A6B5A; /* Secondary accent — deep sage green */
--sage-lt: #7A9E8A; /* Light sage — hover states */
--blush: #D4A9A0; /* Tertiary — warm blush (rare) */
--ink: #1E1A14; /* Primary text — near-black warm */
--ink-soft: #4A4438; /* Body text — warm dark brown */
--ink-mute: #8A8070; /* Muted text — labels, captions */

/* ■■ Typography ■■ */
--f-serif: var(--font-cormorant), 'Georgia', serif; /* Cormorant Garamond */
--f-sans: var(--font-jost), 'Helvetica Neue', sans-serif; /* Jost */
}

/* ■■ Logo Mark ■■ */
.logo-dot {
width: 8px; height: 8px; border-radius: 50%;
background: var(--sage); /* Pulsing sage dot — logo element */
animation: pulse 2.8s ease infinite;
}
@keyframes pulse {
0%,100% { box-shadow: 0 0 0 0 rgba(74,107,90,0.4); }
50% { box-shadow: 0 0 0 6px rgba(74,107,90,0); }
}

/* ■■ 3D Scene ■■ */
.scene-root { perspective: 1200px; perspective-origin: 50% 40%; }

/* ■■ Kinetic Headline ■■ */
.headline { font-size: clamp(56px, 10vw, 108px); font-weight: 300;
line-height: 1.05; letter-spacing: -0.015em; font-family: var(--f-serif); }
.word-italic-sage { font-style: italic; color: var(--sage); }
.word-italic-gold { font-style: italic; color: var(--gold); }

/* ■■ Pillar Tags ■■ */
.pillar { border: 1px solid rgba(160,133,88,0.25); border-radius: 2px;
background: rgba(244,239,228,0.6); backdrop-filter: blur(4px); }
```

5.10 Build Configuration & Domain Record

Listing 5.10 — next.config.js + CNAME (production, verbatim)

```
/* FILE: next.config.js | © 2026 HealthLedger AI — Maria Polychroniadou */
```

```
/* DESCRIPTION: Next.js 14 build configuration */

const nextConfig = {
output: 'export', // Static HTML export — deployed to healthledgerai.com
trailingSlash: true, // Clean URLs for static hosting
images: { unoptimized: true },
eslint: { ignoreDuringBuilds: true },
typescript: { ignoreBuildErrors: true },
};
module.exports = nextConfig;

/* FILE: CNAME (repository root) */
/* Domain registration record for GitHub Pages / static hosting */
healthledgerai.com
```

6. PRIOR ART ANALYSIS

The following analysis identifies the closest known prior art and explains how the present invention distinguishes itself. This section is provided to assist counsel in assessing novelty and non-obviousness.

General-Purpose AI Assistants (OpenAI ChatGPT, Google Gemini, Anthropic Claude)

Distinction: These systems are trained on general internet data and possess no knowledge of hospital cooperation agreements, institution-specific charts of accounts, or healthcare accounting conventions. They cannot produce clause-cited, auditable compliance determinations. They do not implement the RAG pipeline, managed agent architecture, or on-site immersion methodology of the present invention.

Legal Contract AI (e.g. Harvey AI, Luminance, Kira Systems)

Distinction: These systems focus on legal contract review and classification, not on accounting compliance or the translation of contractual clauses into specific accounting entries. They do not target hospital accounting departments, do not use institution-specific charts of accounts, and do not provide real-time point-of-decision accounting guidance.

Healthcare ERP Systems (SAP S/4HANA Healthcare, Oracle Health)

Distinction: ERP systems provide structured accounting workflows but do not perform natural-language compliance reasoning, do not ingest unstructured cooperation agreement text, and do not produce clause-traceable compliance guidance. They require manual configuration of all rules by IT staff.

Document Management / Contract Lifecycle Management (DocuSign CLM, Icertis)

Distinction: These systems store and search contracts but do not provide accounting-specific compliance reasoning, domain-adapted AI models, or on-site knowledge capture.

General RAG Systems

Distinction: While retrieval-augmented generation is a known technique, the present invention distinguishes itself through: (a) the healthcare accounting-specific clause taxonomy; (b) the institution-specific managed agent architecture; (c) the on-site immersion methodology as the training data source; and (d) the specific combination producing clause-traceable accounting compliance determinations — a combination not found in any known prior art.

Non-Obviousness Argument: The combination of (i) on-site immersion as a training data acquisition methodology for domain-specific AI; (ii) the six-category healthcare accounting clause taxonomy; (iii) the institution-specific managed agent pattern; and (iv) the clause-traceable RAG compliance output — is not suggested by any known prior art, individually or in combination. The on-site immersion methodology in particular represents a wholly novel approach to domain adaptation that is both non-obvious and central to the system's effectiveness.

7. DATE OF CONCEPTION & REDUCTION TO PRACTICE

Conception Date:	**2026 (documented in doctoral research; exact date to be confirmed by inventor)**
Reduction to Practice:	**2026 — working system implemented; website live at healthledgerai.com**
First Public Disclosure:	**Not yet made (pre-filing; no public disclosure to date)**
Priority Date Target:	**File provisional application as soon as possible to establish priority**
Evidence of Conception:	**Git repository history (mrply97/AI on GitHub); CNAME record for healthledgerai.com; source code timestamps; doctoral research documentation**
NOTE TO COUNSEL:	**Inventor should confirm exact date of first conception in writing and provide access to doctoral research documentation as corroborating evidence.**

CRITICAL — Attorney Note: If the website healthledgerai.com is or becomes publicly accessible before a patent application is filed, this constitutes a public disclosure. Under US law (AIA) a 12-month grace period applies from first public disclosure by the inventor. Under European patent law (EPC), public disclosure before filing is an absolute bar (no grace period). **File the provisional application before making the website publicly accessible.**

8. DRAFT PATENT CLAIMS

14 draft claims are presented below covering the core AI system (Claims 1–3 independent), the UI innovations (Claims 4–5 independent), and dependent claims adding technical detail. Final scope and language to be determined with counsel.

Claim 1 — Independent: AI Compliance System

A computer-implemented artificial intelligence compliance system for hospital accounting departments, the system comprising: (a) a document ingestion and indexing module configured to: parse cooperation agreements of a healthcare institution; segment parsed content into individual clauses; classify each clause into one of at least five accounting obligation types comprising RevenueRecognition, CostAllocation, ComplianceTrigger, RevenueThreshold, and PenaltyClause; generate a vector embedding for each clause; and store the embeddings in a searchable vector database indexed by clause identifier, agreement identifier, obligation type, and effective date; (b) a persistent AI agent entity created via a managed agent application programming interface, said agent entity having a stable, durably-stored agent identifier and being configured with an institution-specific instruction set comprising the institution's chart of accounts, internal naming convention mappings, and historical interpretive decisions; (c) a retrieval-augmented generation engine configured, upon receipt of an accounting compliance query, to: embed the query; perform approximate nearest-neighbour search over the vector database to retrieve a ranked set of semantically similar clauses; assemble an augmented prompt comprising the retrieved clauses formatted with their clause identifiers and parent agreement identifiers, the user query, and an instruction to cite the source clause; and submit the augmented prompt to a session created against the persistent agent entity; and (d) a compliance output module configured to extract from the agent response a compliance determination, source clause identifiers, parent agreement references, and a confidence indicator, and to write a structured audit log entry comprising the query, determination, citations, and timestamp.

Claim 2 — Independent: Method of Compliance Guidance

A computer-implemented method for generating real-time accounting compliance guidance for a hospital institution, the method comprising: (a) ingesting the institution's cooperation agreements by: parsing agreement documents; segmenting content into clauses; classifying each clause by accounting obligation type; generating vector embeddings; and indexing embeddings with metadata in a vector database; (b) capturing institutional accounting knowledge through on-site immersion with the institution's finance team, comprising: observing accounting decisions involving cooperation agreement clauses; recording each observed decision as an interpretation record comprising a clause reference, the decision taken, the accounting rationale, and the date; and eliciting tacit naming conventions through structured interviews; (c) creating a persistent AI agent entity configured with an institution-specific instruction set assembled from the captured chart of accounts, naming convention mappings, and interpretation records; (d) upon receipt of an accounting compliance query: embedding the query; retrieving the top-K most semantically similar agreement clauses; assembling a retrieval-augmented prompt; creating a session against the persistent agent entity; and generating a streamed compliance determination citing the source clause identifier and parent agreement; and (e) writing an audit log record comprising the query, the compliance determination, and all cited clause and agreement references.

Claim 3 — Independent: On-Site Immersion Methodology

A method for developing a domain-specific artificial intelligence system for healthcare accounting, the method comprising: (a) conducting a sustained on-site engagement of at least four weeks with the accounting department of a target healthcare institution; (b) during said engagement: observing accounting staff interpreting and applying cooperation agreement clauses in real accounting decisions; recording each decision as a structured interpretation record comprising the clause reference, the decision, the rationale provided by the accountant, and the date of decision; and conducting structured interviews to elicit unwritten interpretive conventions and terminology specific to the institution; (c) formalising elicited conventions as a named entity mapping dictionary linking institution-specific terminology to standard accounting terms; (d) assembling an AI agent instruction set from: the institution's chart of accounts; the named entity mapping dictionary; and a curated selection of interpretation records as few-shot examples; (e) deploying the AI agent with said instruction set for institutional compliance use; and (f) iteratively updating the instruction set as new interpretation records are captured post-deployment, without retraining the underlying language model.

Claim 4 — Independent: Particle-Burst Intro Animation Method

A computer-implemented method for presenting an animated entry experience in a web application, the method comprising: (a) generating a set of particles arranged in a unit sphere using cubic-root radial distribution, each particle having a unit-sphere position vector, a size value, a colour selected from a predefined palette, and a twinkle phase; designating a subset of particles as glint particles having an eighth-power sparkle pulse; (b) rendering a first phase during a first time interval comprising rotating the sphere on a Y-axis and applying sinusoidal X-axis tilt to produce organic breathing motion; (c) rendering a burst phase during a second time interval by expanding the sphere radius outward by a factor of at least eight using an easeInOutCubic interpolation; (d) rendering a hold phase during a third time interval maintaining the scattered particle distribution; and (e) rendering a fade phase during a fourth time interval transitioning the overlay opacity from fully opaque to fully transparent using an easeInQuad interpolation, revealing the underlying application interface.

Claim 5 — Independent: Interactive Particle Field with Dual-Mode Physics

A computer-implemented system for rendering an interactive branded particle field on a viewport canvas, the system comprising: (a) a particle generation module producing particles having positions, velocities, sizes, opacity values, sinusoidal phase offsets, and colours selected from a predefined brand colour palette; (b) an organic drift physics module applying per-particle velocity increments using Lissajous-pattern sinusoidal functions of elapsed time and particle phase offset; (c) a mouse-attraction physics module applying velocity increments toward the cursor position for all particles within a defined attraction radius, wherein the attraction force magnitude is proportional to a power-law function of proximity to the cursor; (d) a velocity damping module reducing particle velocities by a constant factor each frame to produce inertial motion; (e) a rendering module drawing each particle with a radial glow corona and solid core; drawing constellation lines between particles within a defined connection distance; and drawing cursor-to-particle lines for particles within the attraction radius.

Claim 6 — Dependent on Claim 1

The system of Claim 1, wherein the persistent AI agent entity is created using the Anthropic Managed Agents API by calling the beta.agents.create() method with a named agent identifier, the target language model identifier claude-opus-4-8, and the institution-specific instruction set, and wherein the resulting agent identifier is stored in a persistent data store for reuse across all subsequent compliance sessions of that institution.

Claim 7 — Dependent on Claim 1

The system of Claim 1, wherein agent responses are delivered to the user interface by token-by-token streaming via server-sent events using the Anthropic beta.sessions.events.stream() API, enabling real-time display of the compliance determination with a first-token latency target below 800 milliseconds.

Claim 8 — Dependent on Claim 1

The system of Claim 1, wherein the institution-specific instruction set of the persistent agent entity is updated incrementally each time a new interpretation record is captured post-deployment by reassembling the instruction string and updating the agent, with such updates taking effect for all subsequent sessions without retraining the underlying language model.

Claim 9 — Dependent on Claim 2

The method of Claim 2, wherein the on-site immersion engagement of step (b) extends for a period of between four and twelve weeks; and wherein interim system outputs are reviewed by accounting staff during said engagement period and feedback is incorporated into the agent instruction set prior to production deployment.

Claim 10 — Dependent on Claim 1

The system of Claim 1, further comprising a web-based user interface implemented in Next.js 14 and deployed as a static export at the domain healthledgerai.com, said interface comprising: a problem-hook display rotating sequentially through at least three problem statement lines at defined time intervals; a three-step method presentation corresponding to the agreement ingestion, domain training, and live compliance steps; and a research programme enrolment contact form directing submissions to the inventor's designated email address.

Claim 11 — Dependent on Claim 4

The method of Claim 4, wherein the first, second, third, and fourth time intervals end at approximately 2,600ms, 5,400ms, 6,400ms, and 7,800ms respectively from the start of the animation; and wherein the dark backdrop colour is #15120E and the brand interface background colour is #F4EFE4.

Claim 12 — Dependent on Claim 5

The system of Claim 5, wherein the brand colour palette consists of: gold (#A08558), gold-light (#C9AA7C), sage (#4A6B5A), sage-light (#7A9E8A), and blush (#D4A9A0); wherein the connection distance is approximately 130 pixels; wherein the mouse attraction radius is approximately 180 pixels; and wherein the attraction force exponent is approximately 1.8.

Claim 13 — Dependent on Claim 1

The system of Claim 1, further comprising a 3D parallax hero interface comprising: a CSS perspective container with perspective value 1200 pixels; spring-animated rotation values driven by cursor position with rotation range ±7° on the Y-axis and ±5° on the X-axis; and child interface elements positioned at differentiated z-translation depths ranging from -160 pixels to +90 pixels to produce differential parallax motion.

Claim 14 — Independent: Use

Use of the system of any one of Claims 1, 6, 7, 8, 10, or 13 for the purpose of reducing financial exposure in hospital accounting operations arising from inconsistent interpretation of cooperation agreement clauses, by providing real-time, clause-cited, institution-specific accounting compliance guidance to hospital finance department staff at the point of accounting decision.

9. ABSTRACT

HealthLedger AI is an artificial intelligence system providing real-time accounting compliance intelligence for hospital finance departments. The system ingests and semantically indexes a hospital network's cooperation agreements using a six-category accounting clause taxonomy and a vector embedding pipeline, creates a persistent AI agent entity configured with institution-specific knowledge acquired through a novel on-site immersion methodology, and deploys a retrieval-augmented compliance interface that delivers auditable, clause-traceable accounting determinations at the point of decision. Secondary inventive contributions include: a four-phase diamond-glitter particle burst entry animation; a spring-physics 3D parallax hero scene; a brand-palette constellation particle field with dual-mode physics; and a dual-spring branded cursor system. The invention addresses the absence of domain-specific AI tooling for hospital cooperation agreement compliance — a gap causing material financial exposure across hospital accounting departments. The brand identity HealthLedger AI, domain healthledgerai.com, and associated taglines are claimed as trademarks. All source code and original SVG artwork are claimed as copyright works of Maria Polychroniadou, 2026.

10. INVENTOR DECLARATION

I, the undersigned, declare that:

■ I am the sole original inventor of all subject matter described in this disclosure, including the AI compliance system, the on-site immersion methodology, all user interface components, the botanical SVG illustration, and the visual design system.

■ The brand name 'HealthLedger AI', the domain healthledgerai.com, the contact email info@healthledgerai.com, and all taglines identified in Section A.1 are my original creations and I claim all trademark rights therein.

■ All source code, artwork, and design assets listed in Section A.3 are original works of authorship created by me, and all copyright subsists in my name as Maria Polychroniadou.

■ All information provided in this disclosure is true and accurate to the best of my knowledge and belief.

■ I have not made any public disclosure, offer for sale, or public use of this invention prior to the date of this document in any manner that would prejudice the filing of a patent application. I understand that I must inform counsel immediately if any such disclosure occurs.

■ I acknowledge my ongoing duty to disclose to patent counsel all prior art, related publications, third-party products, or co-inventors of which I am or become aware.

■ This disclosure has been prepared in good faith for the purpose of initiating patent and trademark prosecution and is submitted under attorney-client privilege.

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Witnessed By:	
Witness Date:	
Notary / Stamp:	

